

TAKAYAMA, Japan

WMO: 476170

Lat: 36.1561N

Lon: 137.2533E

Elev: 1844

StdP: 13.74

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	19.2	11.0	10.3	27.7	13.6	11.7	27.4	12.2	34.9	11.4	34.7	2.6	90	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.2	91.4	73.4	88.9	72.5	86.1	71.6	75.5	85.5	74.4	83.7	73.5	82.1	8.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	131.9	78.1	72.0	126.9	77.3	71.1	122.9	76.8	40.4	85.9	39.4	83.9	38.4	82.4	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.8	10.4	DB	10.5	95.0	3.4	1.8	8.0	96.3	6.0	97.4	4.1	98.4	1.7	99.7	(4)
(5)			WB	9.9	77.3	3.4	1.3	7.4	78.2	5.5	78.9	3.6	79.7	1.1	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	30.2	31.6	39.0	50.2	61.0	68.7	76.0	77.8	69.7	58.2	46.1	35.8	(6)
(7)		DBStd	17.36	4.53	5.38	5.96	6.97	5.04	4.19	4.03	3.79	5.64	5.79	5.87	5.43	(7)
(8)		HDD50	2153	615	516	343	83	2	0	0	0	0	7	147	440	(8)
(9)		HDD65	5139	1080	936	806	445	141	14	0	0	22	222	568	905	(9)
(10)		CDD50	3545	0	0	2	88	342	560	807	863	592	261	29	1	(10)
(11)		CDD65	1057	0	0	0	1	17	125	342	398	163	11	0	0	(11)
(12)		CDH74	7337	0	0	0	31	279	733	2397	2980	873	44	0	0	(12)
(13)		CDH80	2563	0	0	0	3	47	139	913	1229	229	3	0	0	(13)
(14)	Wind	WSAvg	3.7	2.9	3.3	4.0	4.4	4.5	4.0	3.8	3.9	3.5	3.3	3.1	2.9	(14)
(15)	Precipitation	PrecAvg	63.4	3.8	3.5	4.4	4.5	4.8	6.0	8.0	6.9	7.9	5.6	4.0	3.9	(15)
(16)		PrecMax	83.0	7.8	6.5	7.3	10.9	12.7	12.0	17.8	20.7	18.1	17.6	8.5	9.5	(16)
(17)		PrecMin	43.0	1.2	1.7	2.0	1.1	1.9	2.8	1.3	2.0	2.0	1.3	0.7	1.3	(17)
(18)		PrecStd	10.0	1.5	1.4	1.5	2.1	2.4	2.1	4.6	4.1	3.8	3.2	1.7	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	53.6	65.4	78.3	84.9	87.0	94.2	94.8	89.3	78.3	67.3	56.1	(19)
(20)			MCWB	39.6	41.7	48.6	56.2	61.6	67.4	74.0	74.4	71.8	65.1	55.9	48.9	(20)
(21)		2%	DB	42.5	48.0	59.7	72.6	80.4	83.6	91.1	92.1	85.6	74.4	62.9	51.1	(21)
(22)			MCWB	36.9	39.6	46.3	53.9	59.6	66.2	73.4	73.6	70.7	62.7	53.1	45.4	(22)
(23)		5%	DB	39.5	43.7	55.1	68.4	77.0	81.3	88.4	89.8	82.5	71.3	59.1	47.5	(23)
(24)			MCWB	35.1	37.4	44.5	52.7	59.1	65.4	72.5	73.0	69.4	60.6	52.2	43.3	(24)
(25)		10%	DB	36.8	40.2	50.7	64.1	73.5	78.5	85.5	87.2	79.3	67.9	56.0	44.0	(25)
(26)			MCWB	33.6	35.5	42.4	51.6	57.7	65.2	72.0	72.7	68.3	59.4	51.2	40.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.7	44.6	53.1	62.9	67.5	72.9	77.1	77.0	74.8	68.9	59.5	51.5	(27)
(28)			MCD B	44.9	50.4	57.9	65.6	72.9	79.0	87.6	88.2	83.2	74.0	63.0	53.2	(28)
(29)		2%	WB	37.6	41.0	49.1	58.3	64.8	71.2	75.5	75.6	72.8	65.9	56.3	47.0	(29)
(30)			MCD B	40.8	45.6	56.1	65.8	72.1	76.9	85.6	85.8	80.4	70.7	59.2	50.0	(30)
(31)		5%	WB	35.7	38.4	46.3	55.9	62.6	69.6	74.3	74.7	71.6	63.4	53.9	43.4	(31)
(32)			MCD B	38.5	42.3	52.3	63.7	71.1	75.1	83.7	84.3	78.5	68.3	57.7	46.9	(32)
(33)		10%	WB	34.1	36.1	43.4	53.6	60.7	67.8	73.2	73.9	70.3	61.0	51.5	40.6	(33)
(34)			MCD B	36.2	39.6	49.5	61.3	69.0	74.1	81.8	82.9	76.5	66.1	55.5	43.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	16.1	19.3	23.0	22.9	18.8	16.9	18.2	17.0	18.1	16.8	13.0	(35)
(36)			MCDBR	16.5	21.9	28.1	31.0	30.0	23.7	22.0	22.2	20.5	21.5	21.9	18.7	(36)
(37)			MCWBR	11.3	14.0	15.5	14.1	12.2	8.9	7.2	7.0	7.5	10.2	13.0	12.5	(37)
(38)		5% WB	MCD BR	14.8	19.1	23.1	22.2	22.0	16.0	19.2	18.9	17.0	16.6	17.8	17.1	(38)
(39)			MCWBR	10.6	13.2	15.2	12.0	11.1	7.5	6.9	6.7	7.2	9.0	11.9	12.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.373	0.410	0.481	0.514	0.520	0.547	0.529	0.522	0.462	0.420	0.389	0.371	(40)	
(41)		taud	1.981	1.901	1.833	1.871	1.919	1.922	2.044	2.063	2.187	2.214	2.207	2.100	(41)	
(42)		Ebn at Noon	247	254	247	247	248	240	244	243	252	251	244	241	(42)	
(43)		Edh at Noon	43	53	62	63	61	61	53	51	43	38	35	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	616	864	1169	1460	1661	1533	1499	1540	1228	1006	716	535	(44)	
(45)		RadStd	57	61	89	126	183	117	224	159	100	92	60	60	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page